

possible throughput through parallel processing.

Demand for more flexible test systems

With recent improvements in embedded systems technology and software engineering, it is now possible to greatly influence a device's functionality by just changing the software embedded in it. Software-defined functionality is generally more flexible and scalable, and makes it possible to change the characteristics of devices quite rapidly.

This puts test engineers in a fix because traditional test instruments cannot keep pace with the changes in the device-under-test (DUT) because of their fixed user interfaces and hard-coded functions, which are difficult to change. Hence test engineers are also turning to techniques like software-defined and field-programmable gate array (FPGA)-enabled instrumentation, so they can quickly customise their equipment to meet specific application needs and integrate testing directly into the design process.

Software-based instrumentation comes at two levels—virtual and synthetic. Virtual instrumentation is a combination of software and modular hardware instruments that can be customised by the users for their specific needs. It has been used for over two decades but became very popular in the last few years.

"Almost all of the testing industry has accepted the concept of virtual instrumentation, which equals or exceeds traditional instruments in terms of data rate, flexibility and scalability, with reduced system cost," says Dr Agrawal.

Major applications and emerging trends

The major application areas for T&M equipment are:

1. Communications: Cellular tests, wireless and wire-line tests, and optical tests
2. Aerospace and defence: Electronic warfare, radar, satellite test and signal monitoring/intelligence
3. Research and development: Digital, radio frequency (RF), microwave, etc
4. Manufacturing: In-circuit tests, functional/automated manufacturing tests
5. Security: Surveillance, military communications
6. Core electronics: Data acquisition, control, automation, semiconductor/component test, nanotechnology, RF and microwave

One of the most challenging applications is in the strategic electronics sector comprising military and aerospace electronics. The reliability and repeatability needs of these markets are unmatched as the margin for errors is minimal.

Technological advancements that have a far-reaching impact on the way T&M is being utilised today include:

1. Faster and smaller semiconductor chips
2. Better and brighter displays
3. Convergence or consolidation

Not only have these advancements helped bring down the test-times and cost of T&M instruments—making them faster and affordable—these have also helped in making them easy to use and more efficient. These advancements have triggered T&M to drive the following key customer needs:

1. Take complexity out of test and save time to focus on where it matters the most
2. Provide reliability and repeatability—the key expectations from any test platform today
3. Offer new alternatives for testing, like instruments with multiple capabilities and automated test set-ups

—Arun Dogra, country manager & VP (sales, marketing & support),
Agilent Technologies India

In virtual instrumentation, the software is very powerful—to the extent that it can transform common hardware components into test instruments. For example, an analogue-to-digital converter can be made to function as a virtual oscilloscope. So a combination of common hardware and modular test instruments can be strung together with powerful software to make a test system with completely user-defined functionality. And this can be changed any number of times to cater to the changes in the DUT.

A synthetic instrument on the other hand is purely software-defined. It performs a specific synthesis, analysis, or measurement function on com-

pletely generic, measurement-agnostic hardware. So it is a system that can run completely on a PC, without even needing basic and modular T&M instruments. Another recent trend is the use of highly modular and reusable software code, which increases the flexibility of software-based instrumentation solutions even more.

The need for flexibility has also resulted in the greater use of FPGAs in test devices. There are lots of system-level tools for FPGAs these days, so modular instruments with FPGAs can easily be reprogrammed by engineers according to their needs. In fact, with modern software tools engineers can rapidly configure

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